Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	18 July 2026
Team ID	SWTID-2026-2426
Project Name	Food Ordering Behaviour and Consumer Trends: A Structured Analysis of Choices and Habits
Maximum Marks	4 Marks

Technical Architecture:

The solution follows an analytical / business-intelligence architecture built around Tableau, rather than a traditional client-server web application. Raw food-ordering data (Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City, Time Taken for Delivery) is collected in CSV / Excel / database form, cleaned and pre-processed, connected to Tableau as a live or extracted data source, modelled using calculated fields and table calculations, and visualized through interactive dashboards. The dashboards are then published to Tableau Server / Tableau Cloud so that Business Analysts, Operations Managers, and other stakeholders can access role-based, interactive views of order trends, delivery performance, and customer behaviour.

Data Flow: Data Sources (CSV / Excel / SQL Database) → Data Cleaning & Preprocessing (handling missing values, duplicates, formatting) → Tableau Data Source / Extract → Data Modelling (joins, calculated fields, groups, sets) → Interactive Dashboards & Visualizations (charts, maps, KPI tiles) → Publishing to Tableau Server / Tableau Cloud → End Users (Business Analyst, Operations Manager, Customer-facing insights).

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboards through which users (Business Analyst, Operations Manager, stakeholders) view and explore insights.	Tableau Desktop, Tableau Public, Tableau Web / Browser-based Dashboards
2.	Application Logic-1	Data collection and preprocessing: handling missing values, duplicates, formatting and structuring the raw dataset.	Excel, Python (Pandas), SQL
3.	Application Logic-2	Data modelling and aggregation: calculated fields, table calculations, groups and sets for deriving business metrics.	Tableau Calculated Fields, Tableau Table Calculations
4.	Application Logic-3	Visualization and analysis logic that renders charts, maps, and KPI dashboards from the modelled data.	Tableau Desktop Visualization Engine
5.	Database	Storage of structured order data such as Order ID, Order Value, Delivery Fee, Cuisine, Meal Type, City and Delivery Time.	MySQL / CSV / Excel
6.	Cloud Database	Hosting of published dashboards and data sources for wider, role-based access.	Tableau Cloud, Tableau Server
7.	File Storage	Storage of raw and processed data files used as inputs to Tableau.	Local Filesystem, CSV / Excel Files
8.	External API-1	Not applicable for the current scope of this analytical project.	Not Applicable
9.	External API-2	Not applicable for the current scope of this analytical project.	Not Applicable

S.No	Component	Description	Technology
10.	Machine Learning Model	Not applicable; the project focuses on descriptive and diagnostic analytics rather than predictive modelling.	Not Applicable
11.	Infrastructure (Server / Cloud)	Application deployment for dashboard authoring and sharing. Local Server Configuration: Local desktop with Tableau Desktop installed. Cloud Server Configuration: Tableau Cloud / Tableau Server for publishing and sharing dashboards.	Local Desktop, Tableau Server, Tableau Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source tools used for data preprocessing prior to loading data into Tableau.	Python (Pandas, NumPy)
2.	Security Implementations	Role-based access controls on published dashboards; secure storage and restricted sharing of the underlying dataset.	Tableau Server / Cloud Permissions, Role-Based Access Control (RBAC)
3.	Scalable Architecture	The dashboard and data-source design allows additional cities, cuisines, and time periods of order data to be incorporated without redesign.	Tableau Extracts, Data Source Filters
4.	Availability	Dashboards published on Tableau Cloud / Tableau Server remain accessible to stakeholders on demand from any location.	Tableau Cloud / Tableau Server hosting
5.	Performance	Use of Tableau extracts, filters, and optimized calculated fields to ensure fast rendering of dashboards even with large volumes of order data.	Tableau Extracts, Data Source Filters, Context Filters